

Week 8 & 9: Score, Health, Game Over & Dynamic Zombie Spawning

Activities Performed

- Improved the existing score system.
- Added a health system for the player.
- Implemented collision detection between zombies and the player.
- Reduced player health when a zombie touches the player.
- Added a Game Over condition when player health reaches zero.
- Implemented continuous zombie spawning instead of a fixed number of zombies.
- Added a zombie spawn timer to control the spawning rate.
- Made the zombie spawning speed increase after every 10 points.
- Added a maximum spawning speed to keep the game playable.
- Added a restart option after Game Over.
- Tested the complete gameplay system and fixed basic issues.

Implementation Details

During this phase, the score and health systems were added to the game. The player's health starts at 100. When a zombie collides with the player, the player's health decreases. When the health reaches zero, the game enters the Game Over state.

The zombie system was also improved. Instead of creating only a fixed number of zombies, a spawn timer continuously creates new zombies during gameplay.

Initially, a new zombie appears every 2 seconds. As the player eliminates zombies and reaches every 10 points, the zombie spawn interval becomes shorter. This gradually increases the difficulty of the game.

The spawning speed is limited to a minimum of 0.5 seconds so that the game does not become excessively fast.

Dynamic Zombie Spawning

Score	Spawn Interval
0–9	2.0 seconds
10–19	1.7 seconds
20–29	1.4 seconds
30–39	1.1 seconds
40–49	0.8 seconds
50+	0.5 seconds

This allows the game to start slowly and gradually become more challenging.

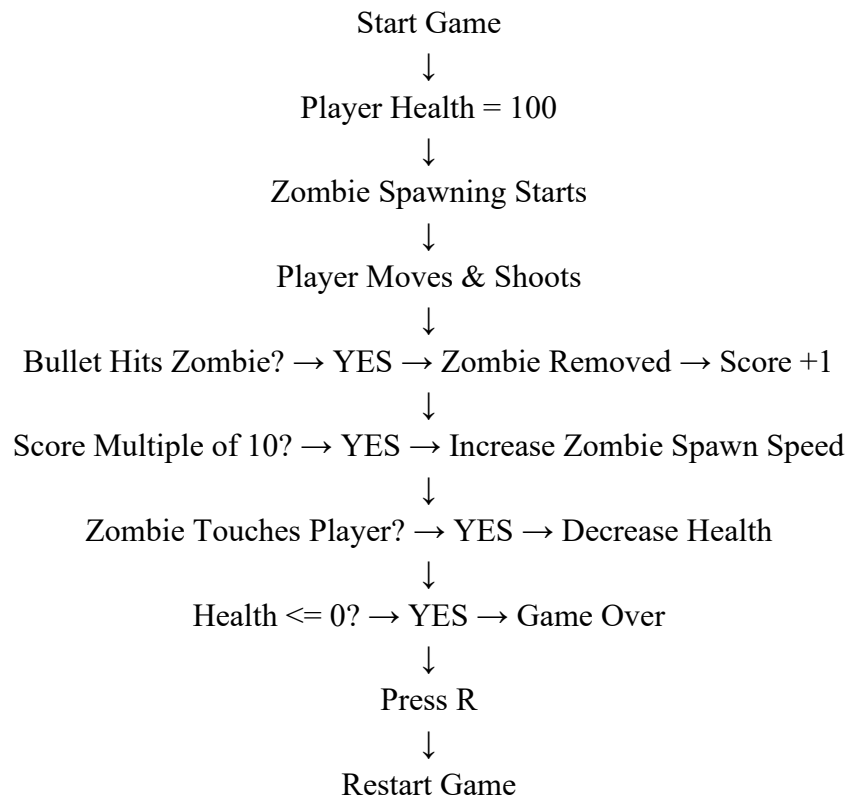
Game Rules

Action	Result
Shoot Zombie	Score increases by 1
Zombie touches Player	Health decreases
Score reaches 10	Zombie spawning becomes faster
Health reaches 0	Game Over
Press R after Game Over	Restart game

Controls

Key	Function
W	Move Up
S	Move Down
A	Move Left
D	Move Right
SPACE	Shoot
R	Restart after Game Over

Game Flow



Outcome

The score, health, Game Over, restart, and dynamic zombie spawning systems were successfully implemented.

The game now continuously generates new zombies instead of stopping after a fixed number of zombies. The difficulty also increases gradually as the player's score increases.

The major gameplay features have now been completed and the game has been tested for basic functionality. The project is ready for the final documentation and presentation phase.

Code Summary

Player.java

- Stores the player's position, health, speed and size.
- Handles player movement.
- Prevents the player from leaving the game screen.
- Provides the player's collision boundary.

Zombie.java

- Stores the zombie's position and speed.
- Moves the zombie toward the player.
- Provides the zombie's collision boundary.

Bullet.java

- Creates and displays bullets.
- Moves bullets across the screen.
- Provides the bullet's collision boundary.

GamePanel.java

- Controls the main game loop.
- Handles player, zombie and bullet movement.
- Detects collisions.
- Updates the score and health.
- Controls continuous zombie spawning.
- Increases spawn speed every 10 points.
- Displays the Game Over screen.
- Handles game restart.